Automated Reconciliation of Radiology Reports with Discharge Summaries

Bevan Koopman, Research Scientist

AUSTRALIAN E-HEALTH RESEARCH CENTRE



@bevan_koopman

The problem of missed limb fractions in the Emergency Room

An obvious fracture





Colles fracture

A subtle abnormality



Defect through cortex

The problem of missed limb fractions in the ER



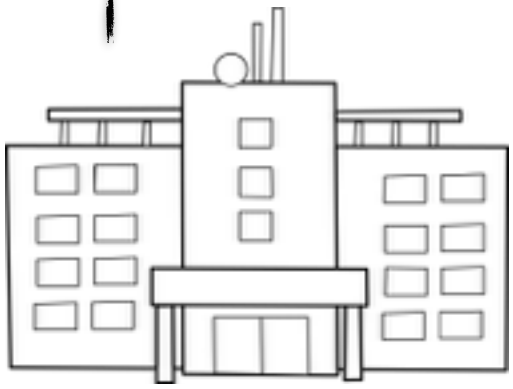
Patient presents to Emergency Room
with limb injury

The problem of missed limb fractions in the ER



Xray reviewed
by ER doctor







Patient presents to Emergency Room
with limb injury

The problem of missed limb fractions in the ER



Xray reviewed
by ER doctor



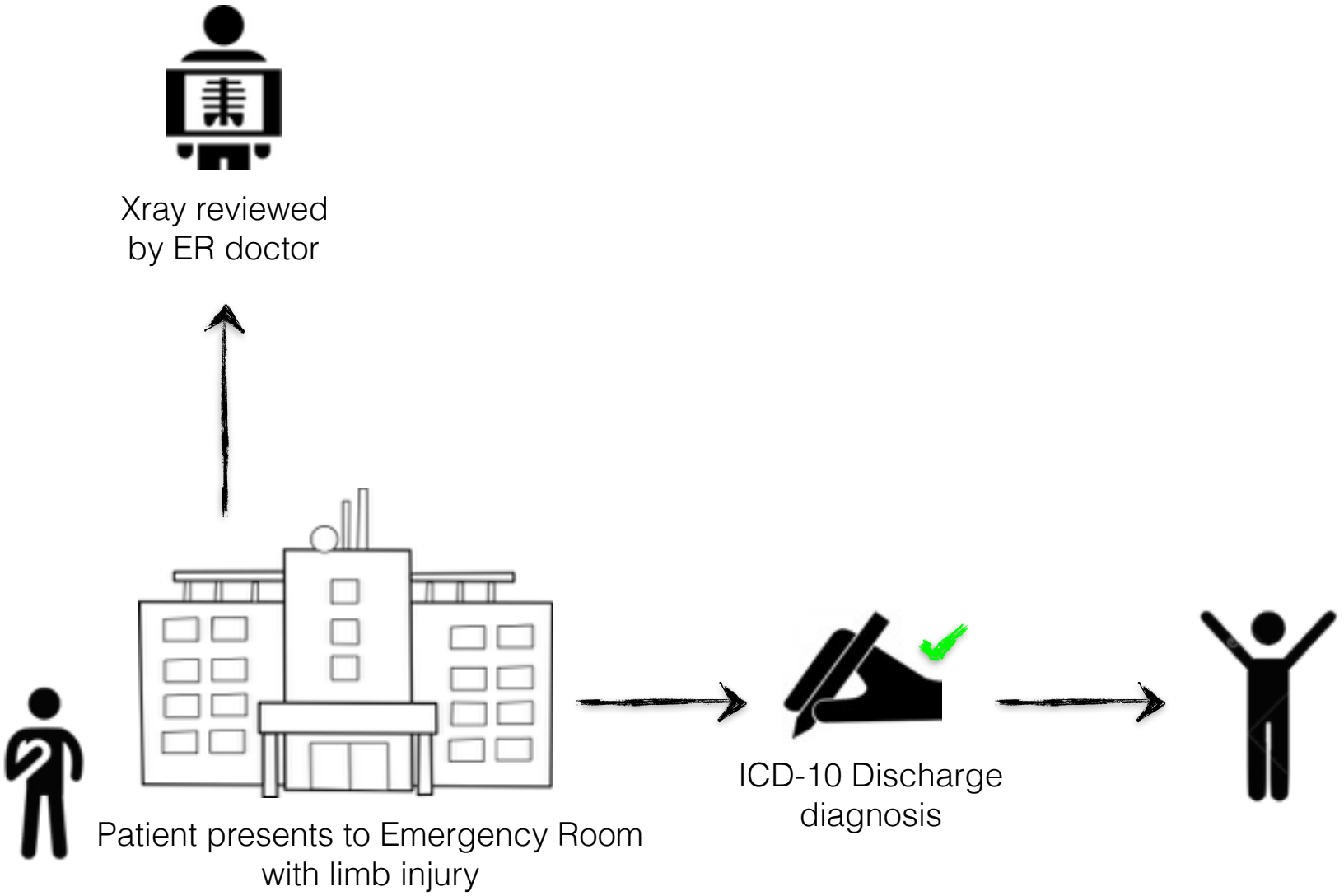


Patient presents to Emergency Room
with limb injury

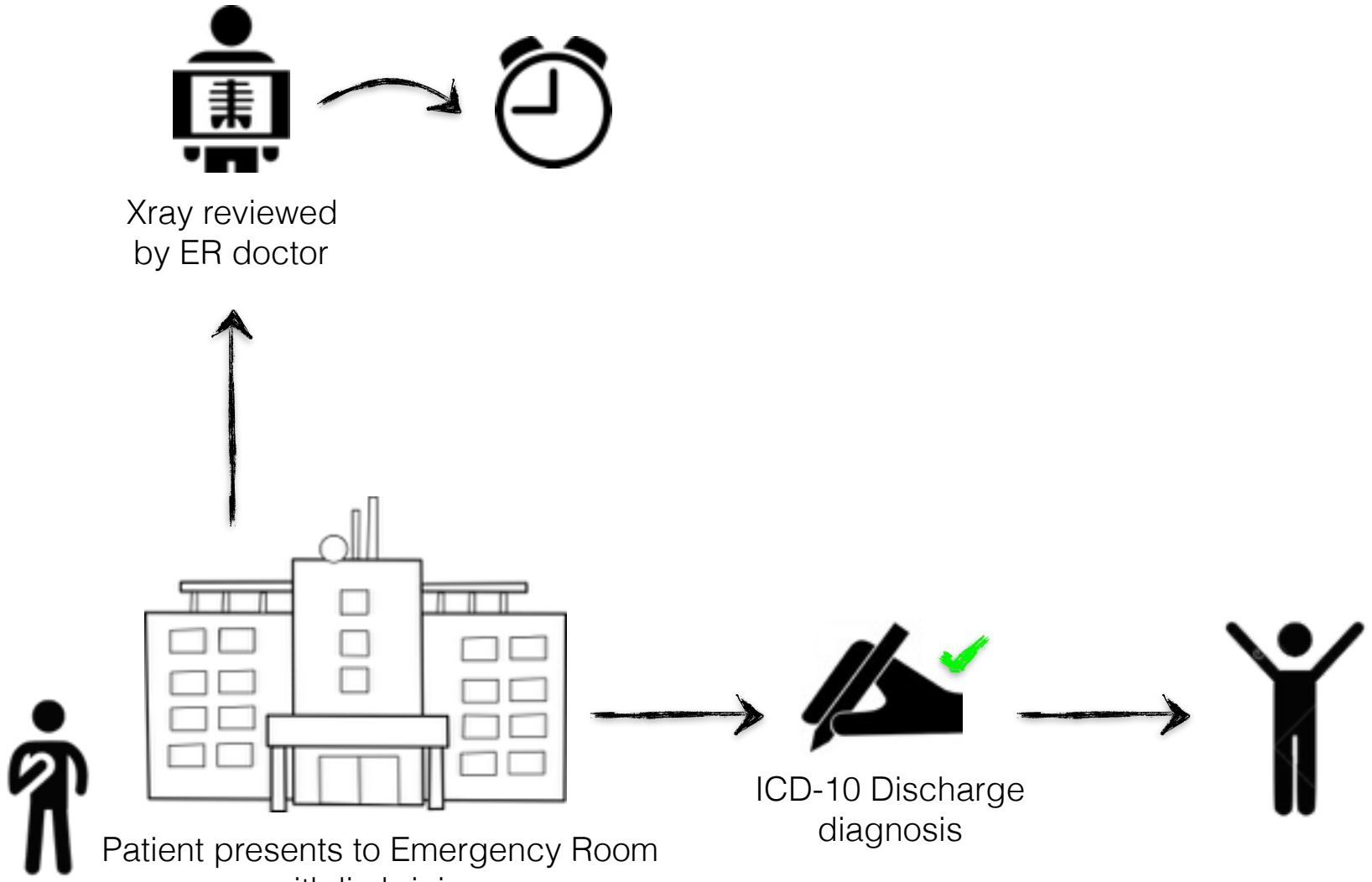


ICD-10 Discharge
diagnosis

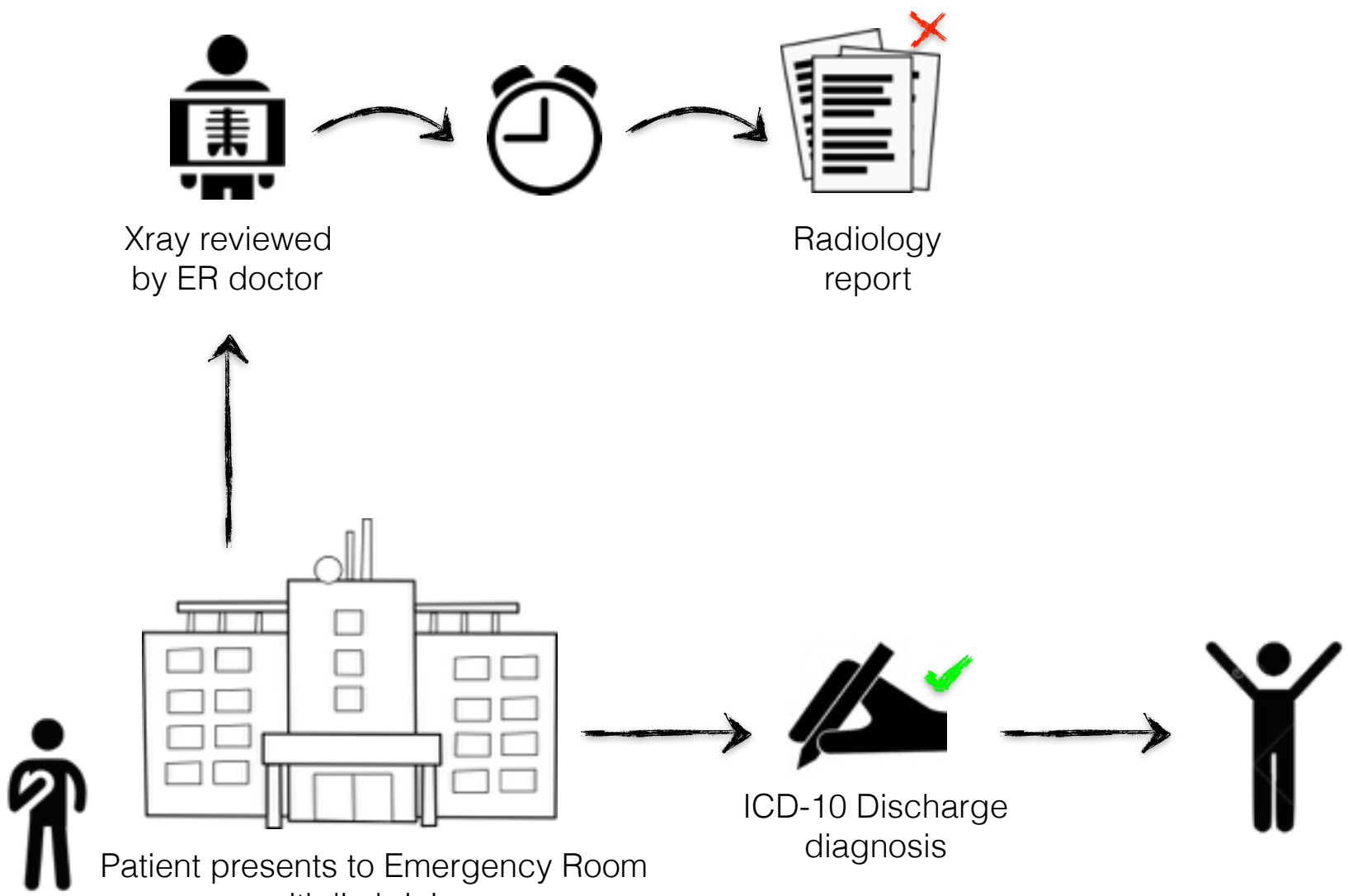
The problem of missed limb fractions in the ER



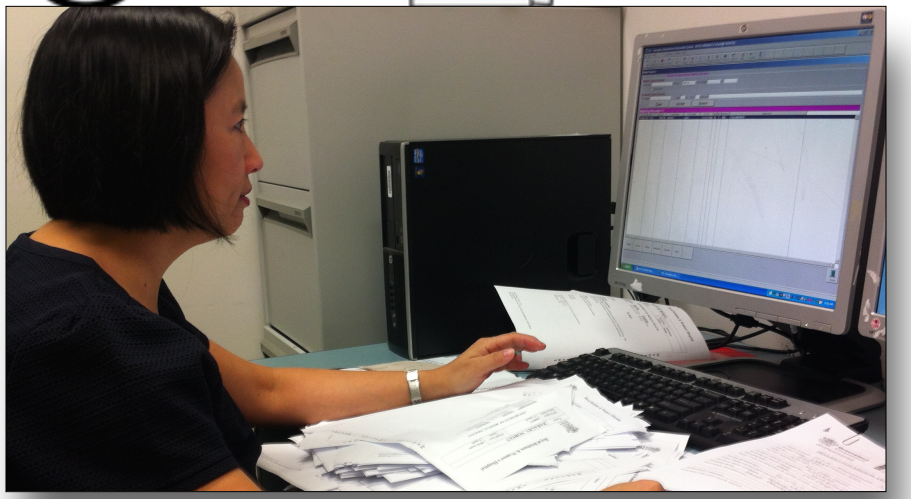
The problem of missed limb fractions in the ER



The problem of missed limb fractions in the ER



The problem of missed limb fractions in the ER

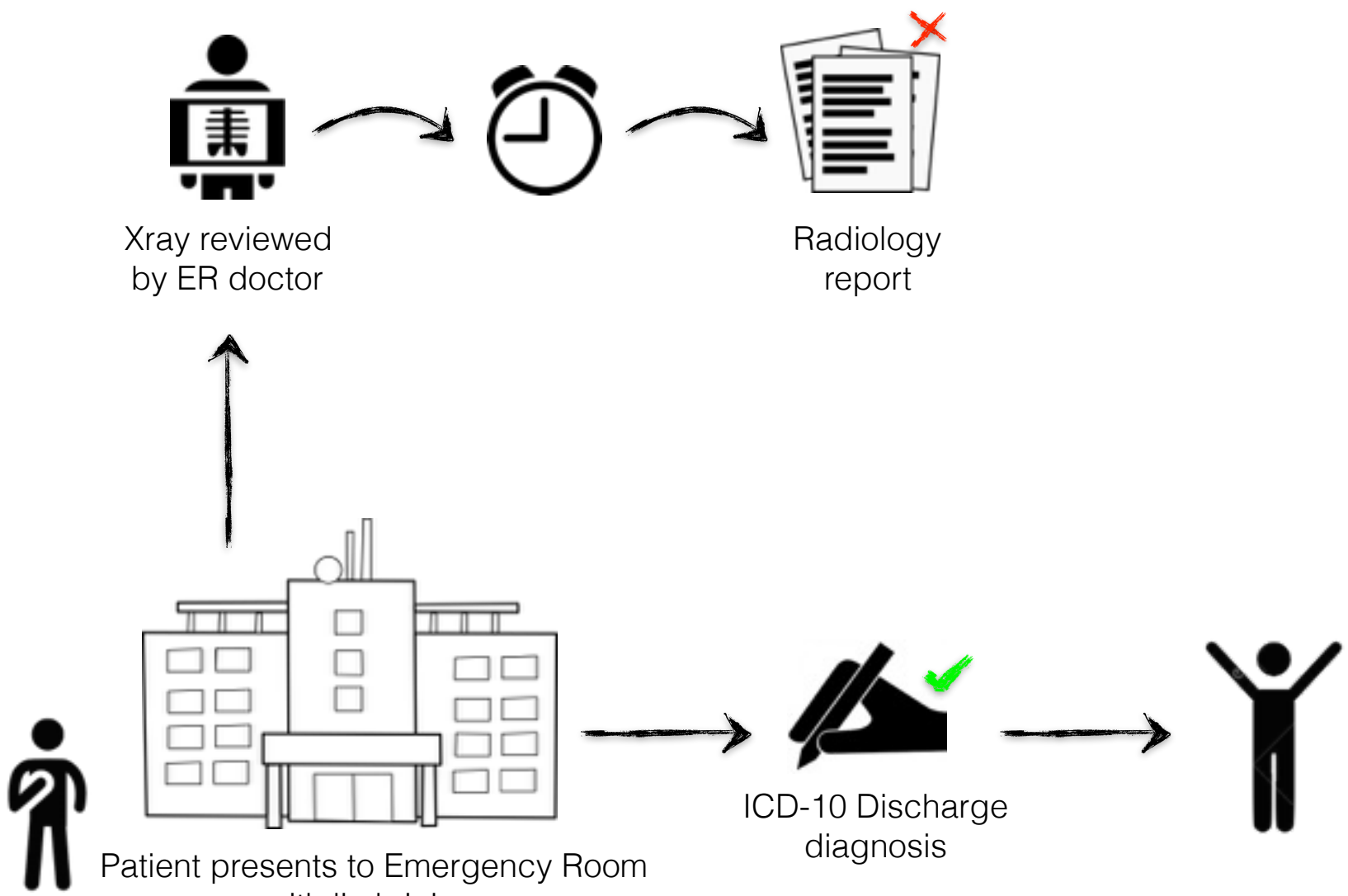


Xray reviewed
by ER doctor

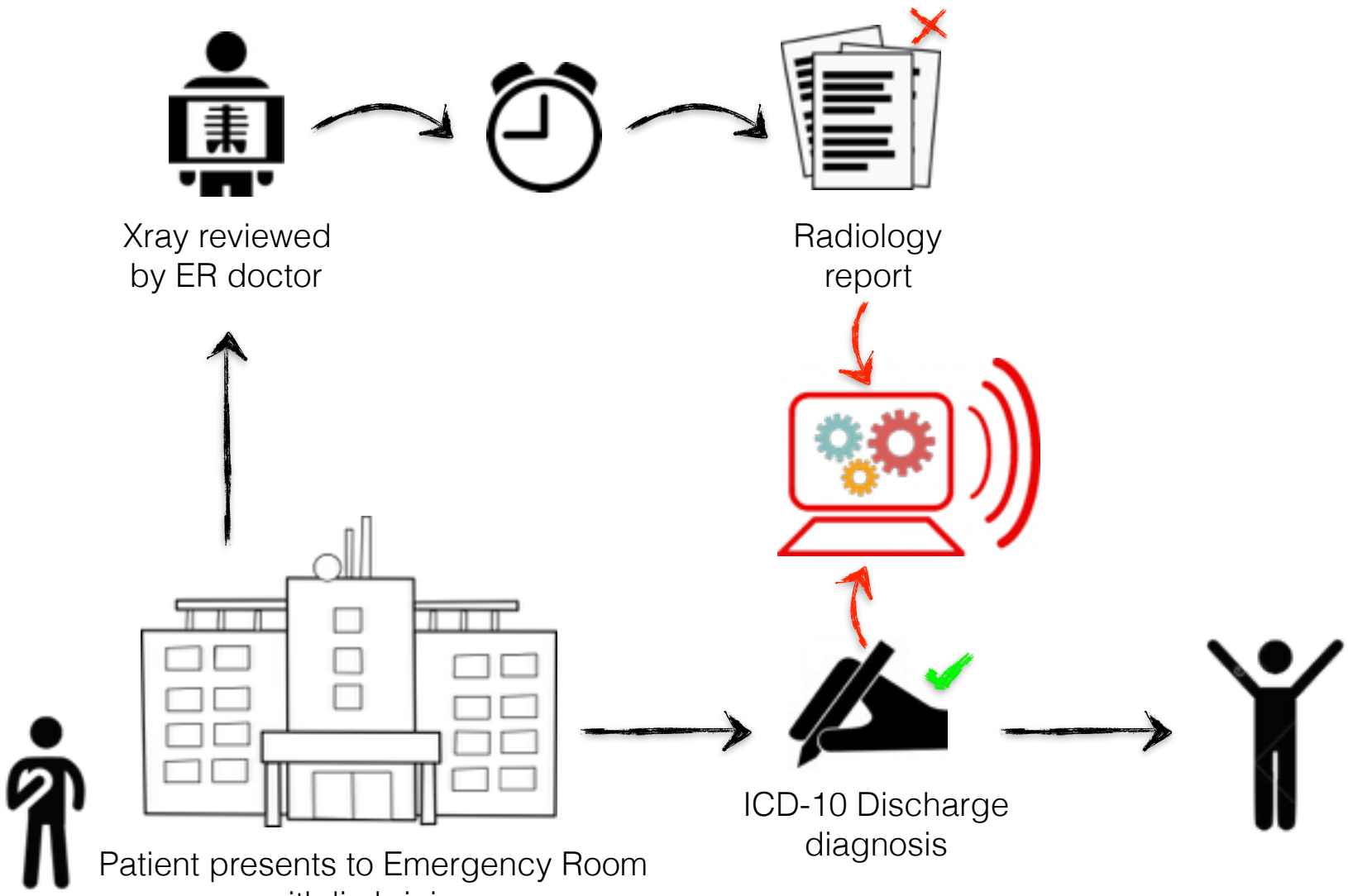
Patient presents to Emergency Room
with limb injury

ICD-10 Discharge
diagnosis

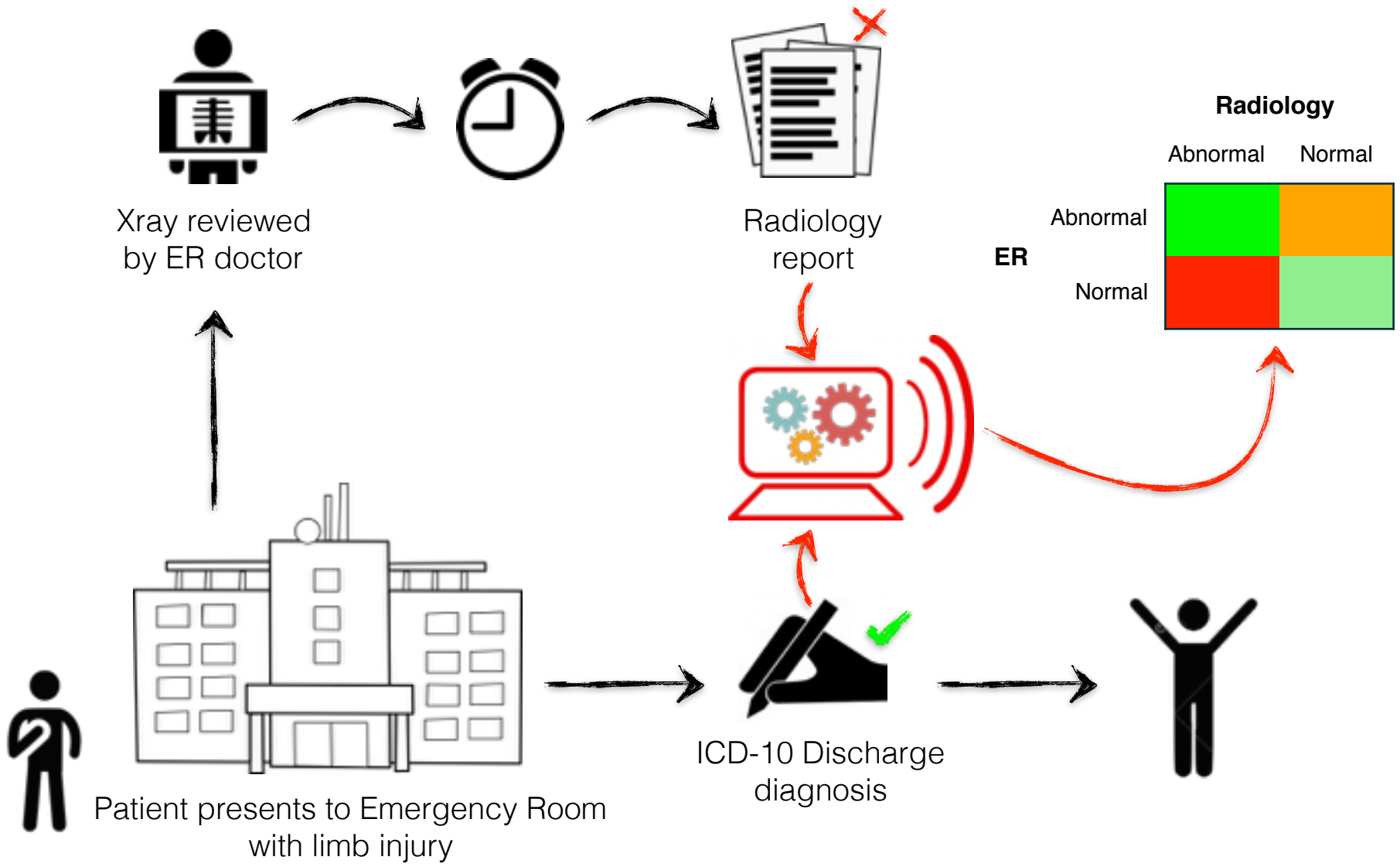
The problem of missed limb fractions in the ER



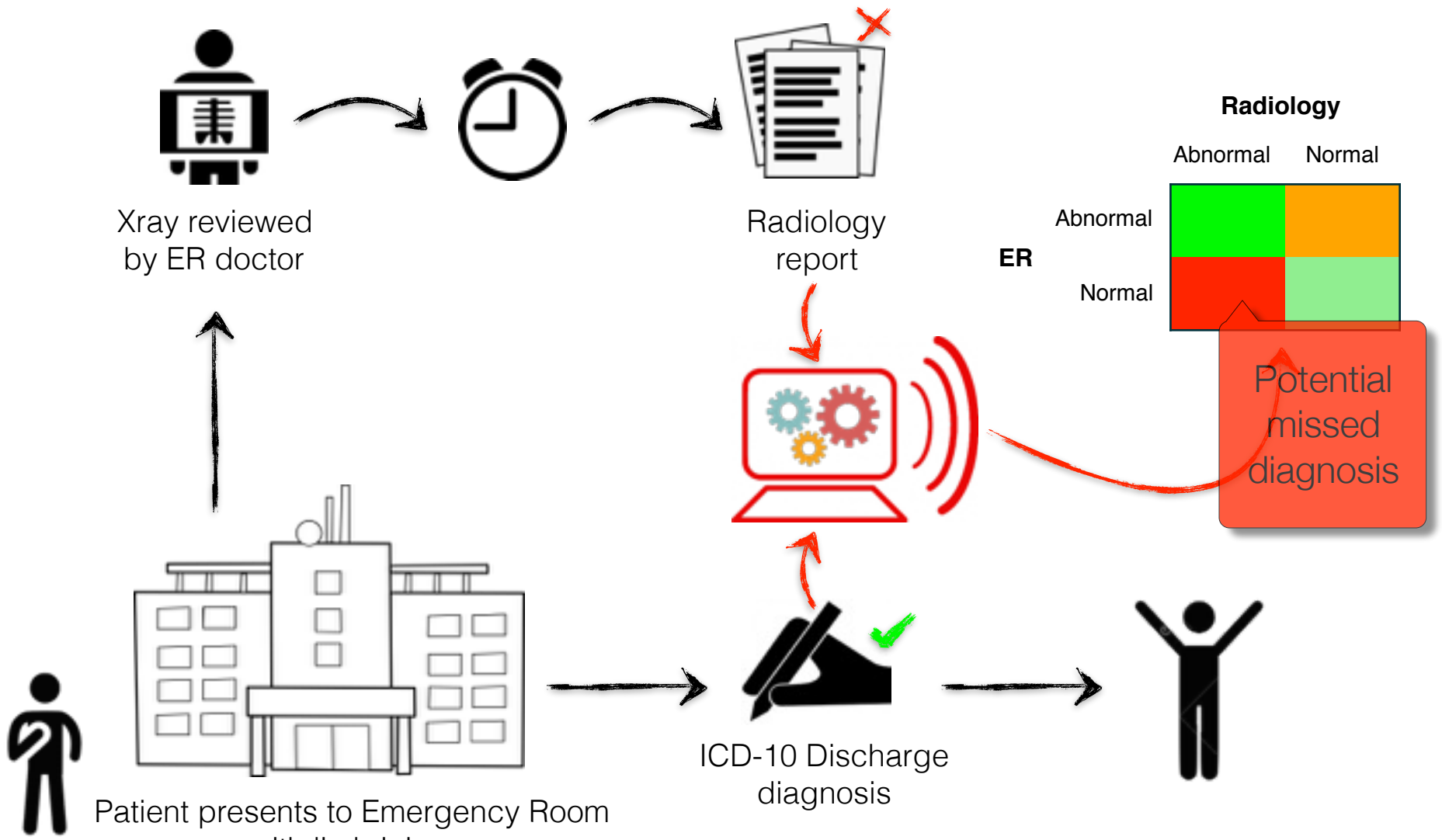
The problem of missed limb fractions in the ER



The problem of missed limb fractions in the ER



The problem of missed limb fractions in the ER



Automatic classification of abnormalities in radiology reports

Three different radiology datasets

Dataset	Description	#Reports	Normal	Abnormal
RBWH	Royal Brisbane & Womens' Hospital (adult)	1,480	58%	42%
RCH	Royal (Brisbane) Childrens' Hospital (child)	498	66%	34%
GCH	Gold Coast Hospital (adult child 38%)	400	62%	38%

Gold Standard by three ER physicians:

- ER Registrar
- Senior Medical Officer
- Adjudicated by Consultant

Extracting features from radiology reports

- Token (i.e., a single word)
- punctuation
- token stem; e.g, “fractur”
- token negation; e.g., “no fractur”
- token stem bi-gram
- token stem tri-gram
- SNOMED-CT concept
- SNOMED-CT concept bi-gram

Train Support Vector Machine (SVM)

Train Support Vector Machine (SVM)

- How effective is a classifier trained on a mixture of hospital reports?

Train Support Vector Machine (SVM)

- How effective is a classifier trained on a mixture of hospital reports?
- How does effectiveness differ when we consider each hospital dataset in isolation?

Train Support Vector Machine (SVM)

- How effective is a classifier trained on a mixture of hospital reports?
- How does effectiveness differ when we consider each hospital dataset in isolation?
- How robust are the methods across different hospitals (train one, test on another)?

Train Support Vector Machine (SVM)

- How effective is a classifier trained on a mixture of hospital reports?
- How does effectiveness differ when we consider each hospital dataset in isolation?
- How robust are the methods across different hospitals (train one, test on another)?
- In comparison with keyword spotting baseline.

(Thomas et al., 2005; Wagholikar et al., 2013).

Classification Results (F-measure)

Training \ Testing	GCH (CA)	RBWH (A)	RCH (C)	All
(CA) GCH	0.95	0.80	0.88	
(A) RBWH	0.84	0.89	0.85	
(C) RCH	0.87	0.81	0.95	
GCH+RBWH			0.88	
GCH+RCH		0.80		
RBWH+RCH	0.84			
All				0.92

C = Children; A = Adults.

33% avg. improvement over keyword baseline

Classification Results (F-measure)

Training \ Testing	GCH (CA)	RBWH (A)	RCH (C)	All
(CA) GCH	0.95	0.80	0.88	
(A) RBWH	0.84	0.89	0.85	
(C) RCH	0.87	0.81	0.95	
GCH+RBWH			0.88	
GCH+RCH		0.80		
RBWH+RCH	0.84			
All				0.92

C = Children; A = Adults.

33% avg. improvement over keyword baseline

Classification Results (F-measure)

Training \ Testing	GCH (CA)	RBWH (A)	RCH (C)	All
(CA) GCH	0.95	0.80	0.88	
(A) RBWH	0.84	0.89	0.85	
(C) RCH	0.87	0.81	0.95	
GCH+RBWH			0.88	
GCH+RCH		0.80		
RBWH+RCH	0.84			
All				0.92

C = Children; A = Adults.

33% avg. improvement over keyword baseline

Classification Results (F-measure)

Training \ Testing	GCH (CA)	RBWH (A)	RCH (C)	All
(CA) GCH	0.95	0.80	0.88	
(A) RBWH	0.84	0.89	0.85	
(C) RCH	0.87	0.81	0.95	
GCH+RBWH			0.88	
GCH+RCH		0.80		
RBWH+RCH	0.84			
All				0.92

C = Children; A = Adults.

33% avg. improvement over keyword baseline

Classification Results (F-measure)

Training \ Testing	GCH (CA)	RBWH (A)	RCH (C)	All
(CA) GCH	0.95	0.80	0.88	
(A) RBWH	0.84	0.89	0.85	
(C) RCH	0.87	0.81	0.95	
GCH+RBWH			0.88	
GCH+RCH		0.80		
RBWH+RCH	0.84			
All				0.92

C = Children; A = Adults.

33% avg. improvement over keyword baseline

Classification Results (F-measure)

Able to identify limb abnormalities with high degree of accuracy.

Robust across different hospital reports.

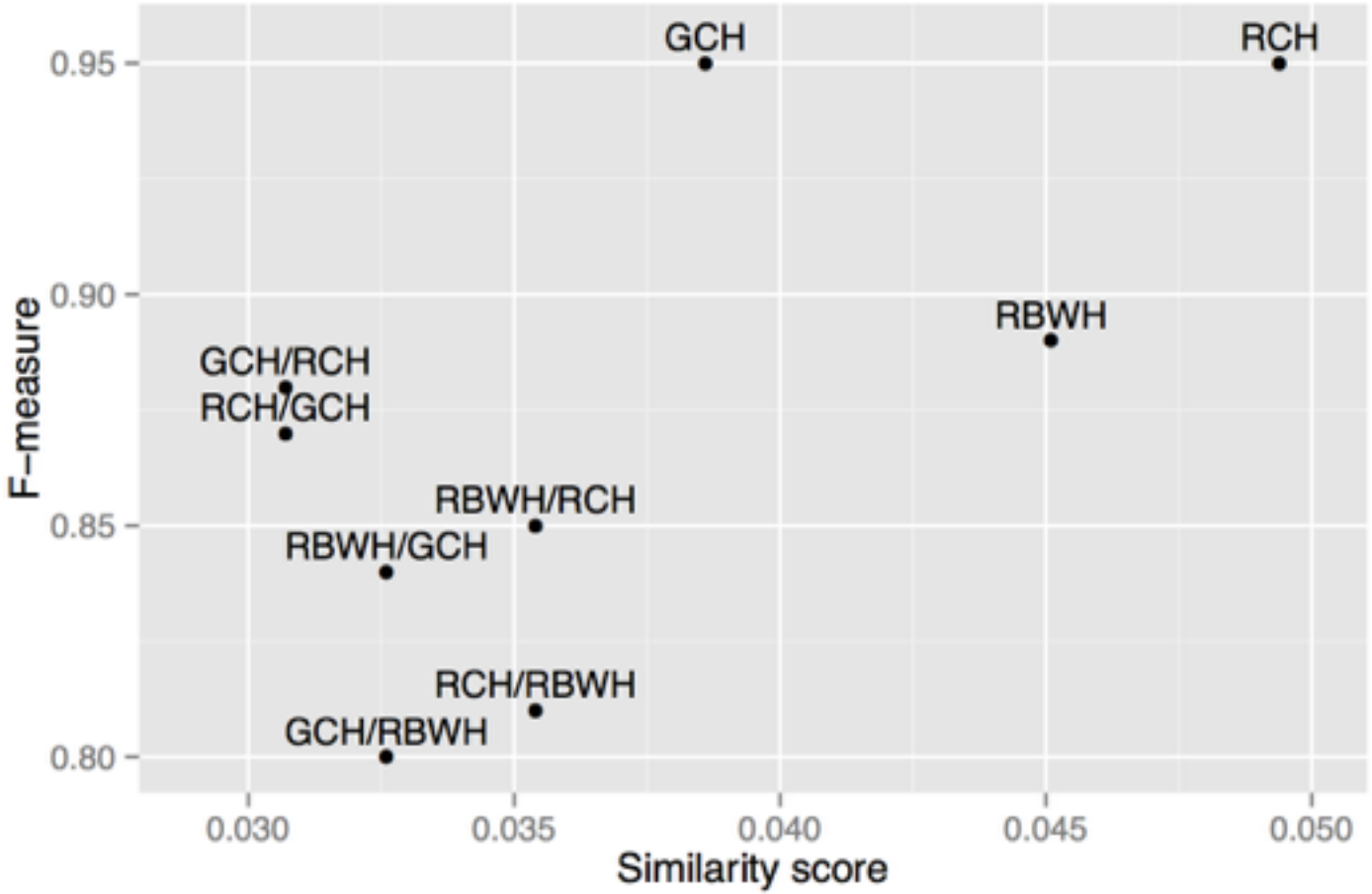
More training data does not equate to better effectiveness.

C = Children; A = Adults.

33% avg. improvement over keyword baseline

How does report similarity affect classification results?

How does report similarity affect classification results?



RCH: C

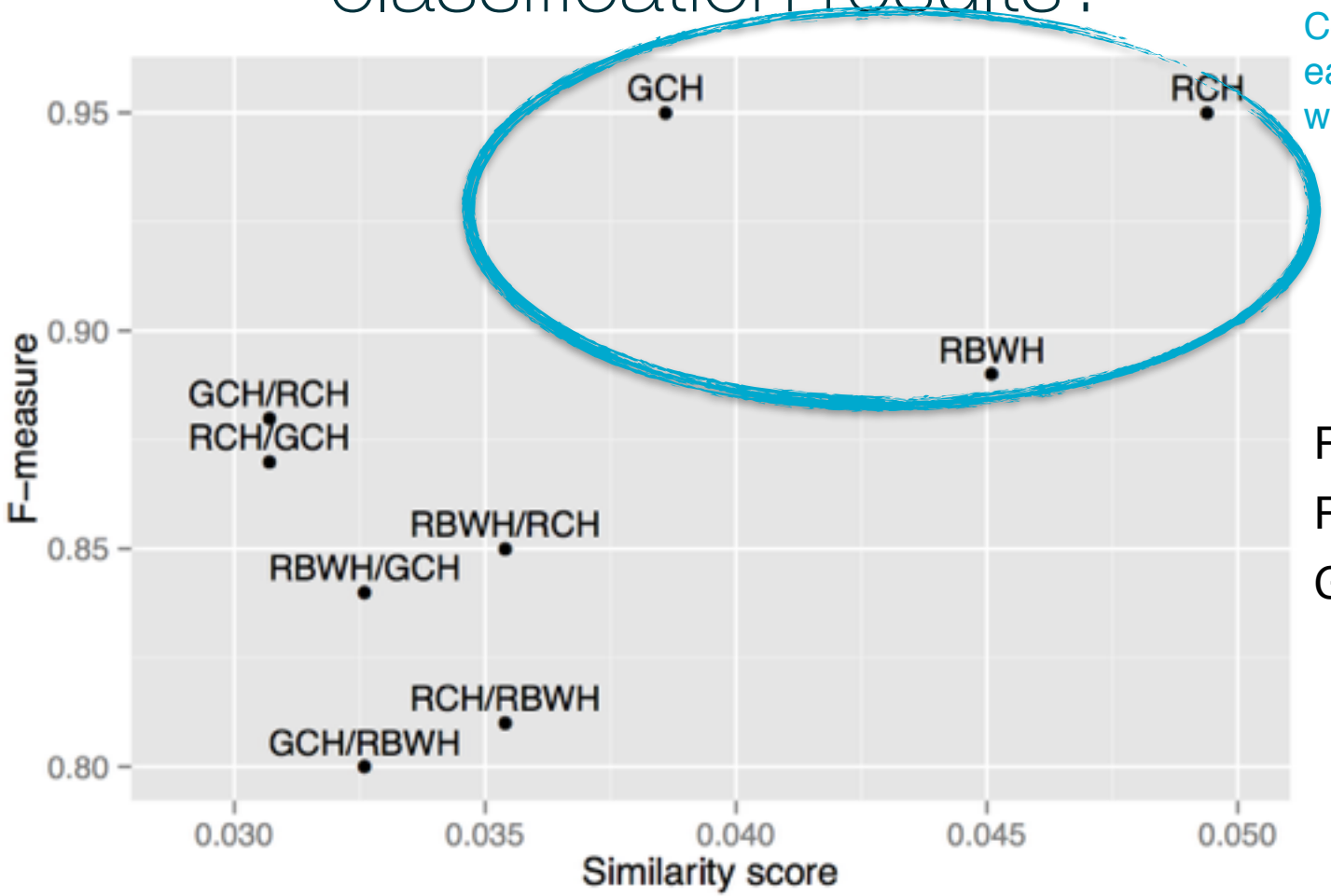
RBWH: A

GCH: C A

Less cohesive

More cohesive

How does report similarity affect classification results?



Cohesive of each dataset with itself.

RCH: C

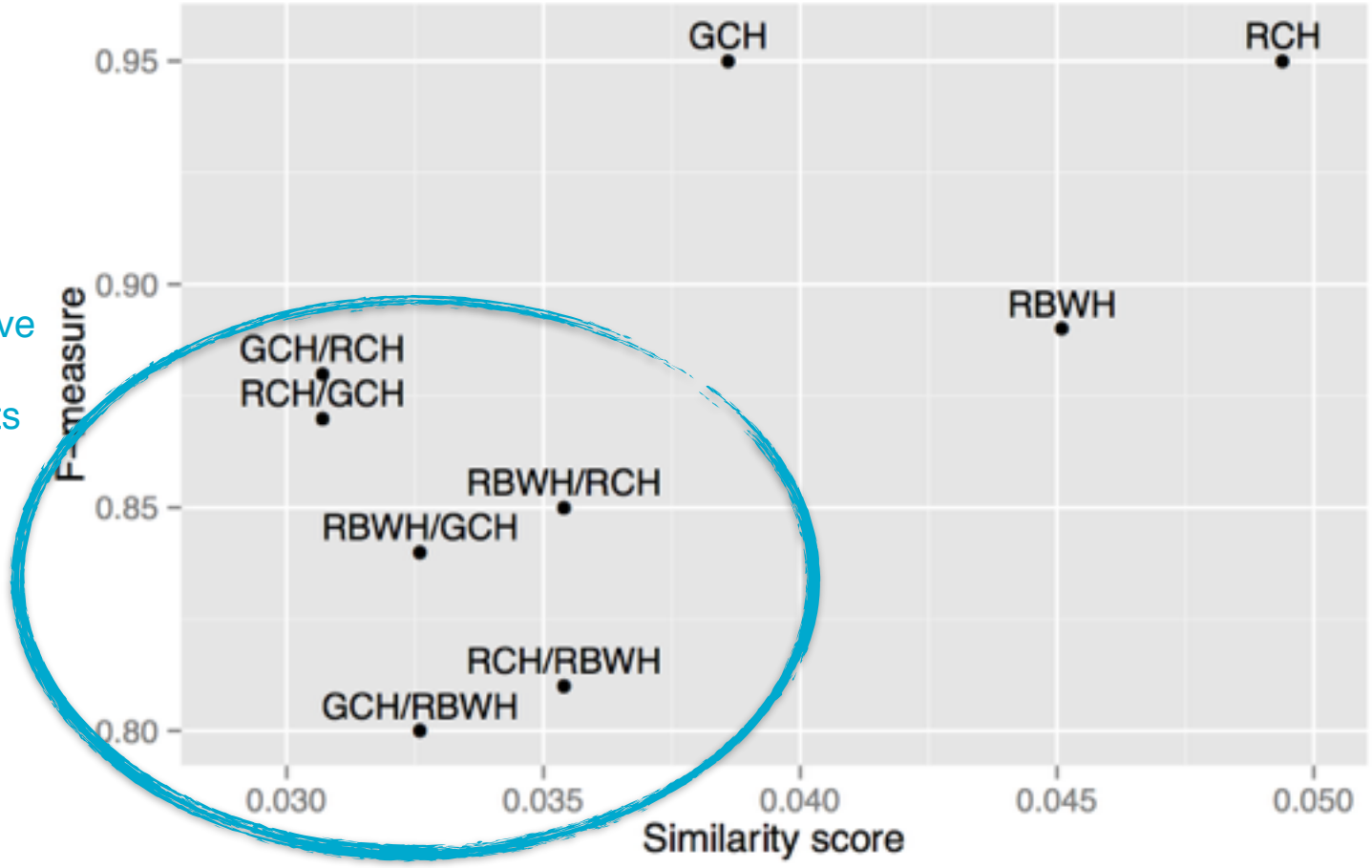
RBWH: A

GCH: C A

Less cohesive

More cohesive

How does report similarity affect classification results?



Cohesive across datasets

Less cohesive

More cohesive

RCH: C

RBWH: A

GCH: C A

How does report similarity affect classification results?

Effectiveness not correlated with similarity.

Variance in training data provides model with a mix of reporting styles.

Overall difference are small - models are robust across different hospitals

Cohesiv
across
datasets

A

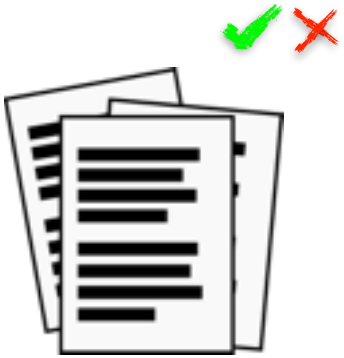
A

Less cohesive

More cohesive

Reconciliation with Discharge Diagnosis

Reconciliation



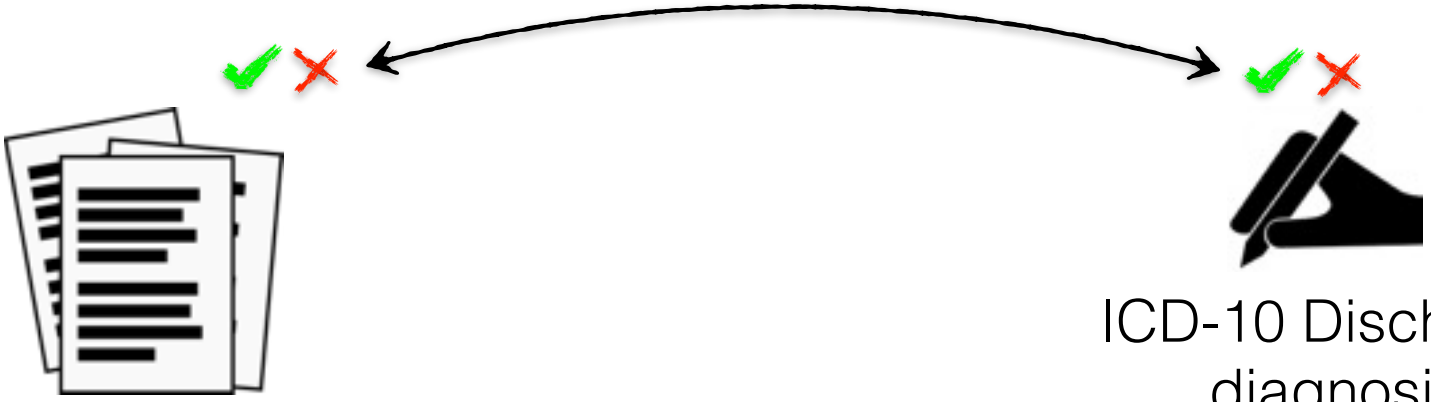
Radiology report classification



ICD-10 Discharge diagnosis

S63.0: Dislocation of wrist
S22.3: Fracture of rib
…

Reconciliation

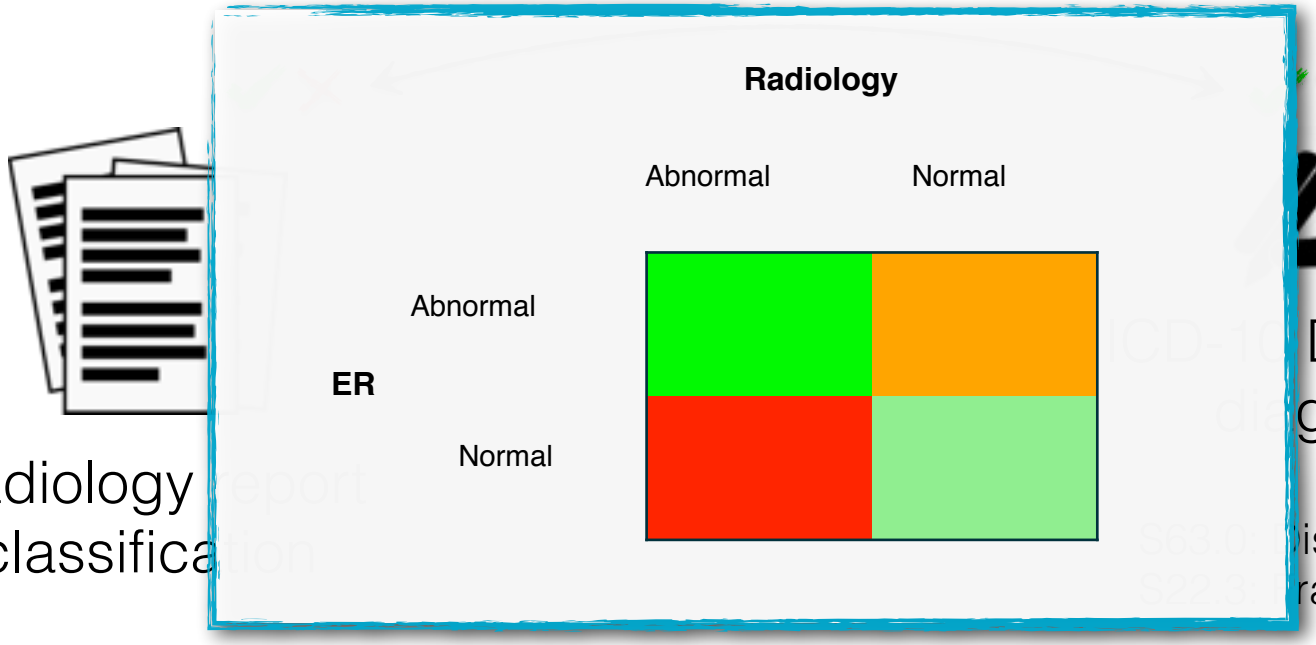


Radiology report classification

ICD-10 Discharge diagnosis

S63.0: Dislocation of wrist
S22.3: Fracture of rib
…

Reconciliation

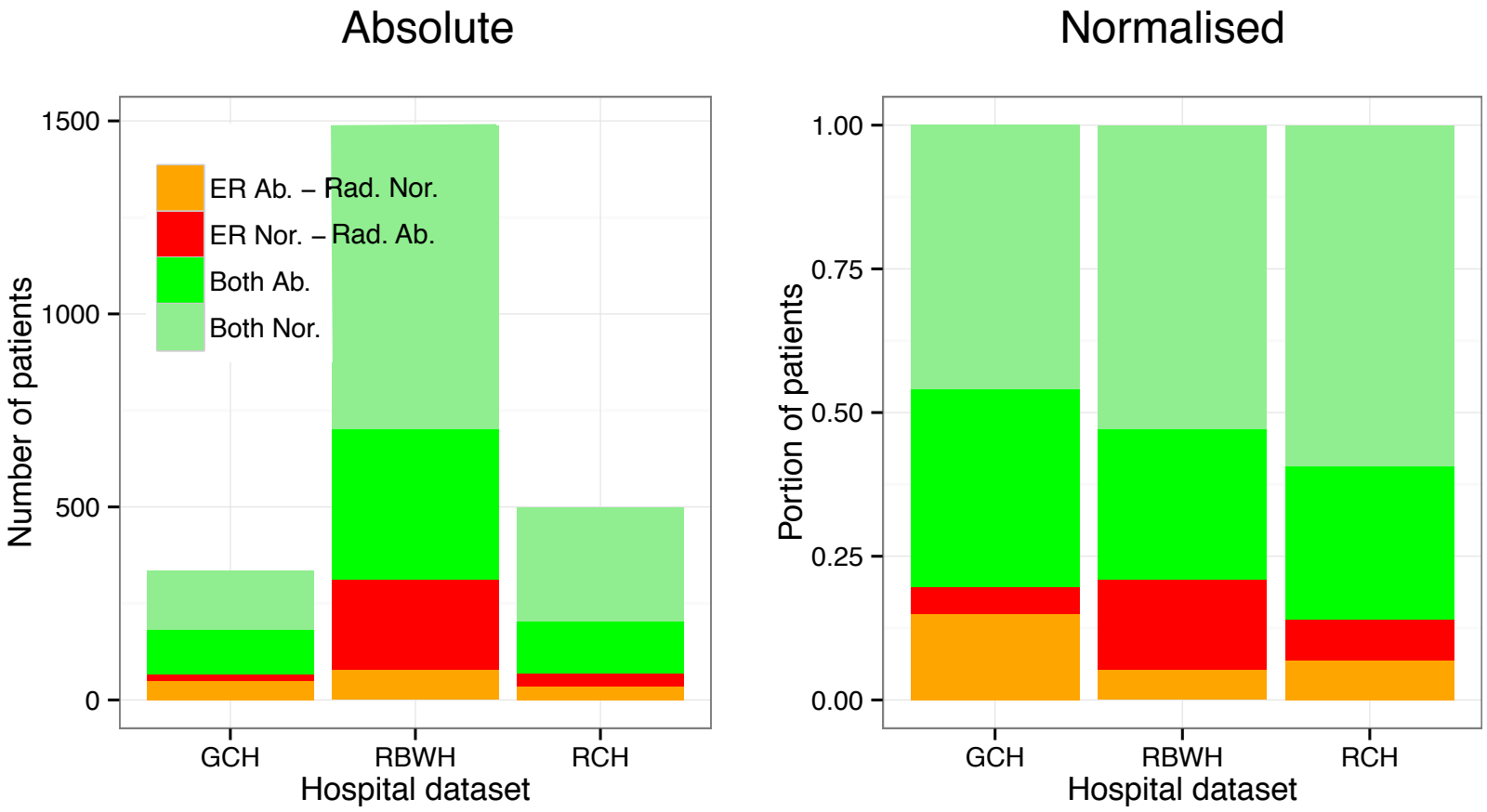


Radiology classification

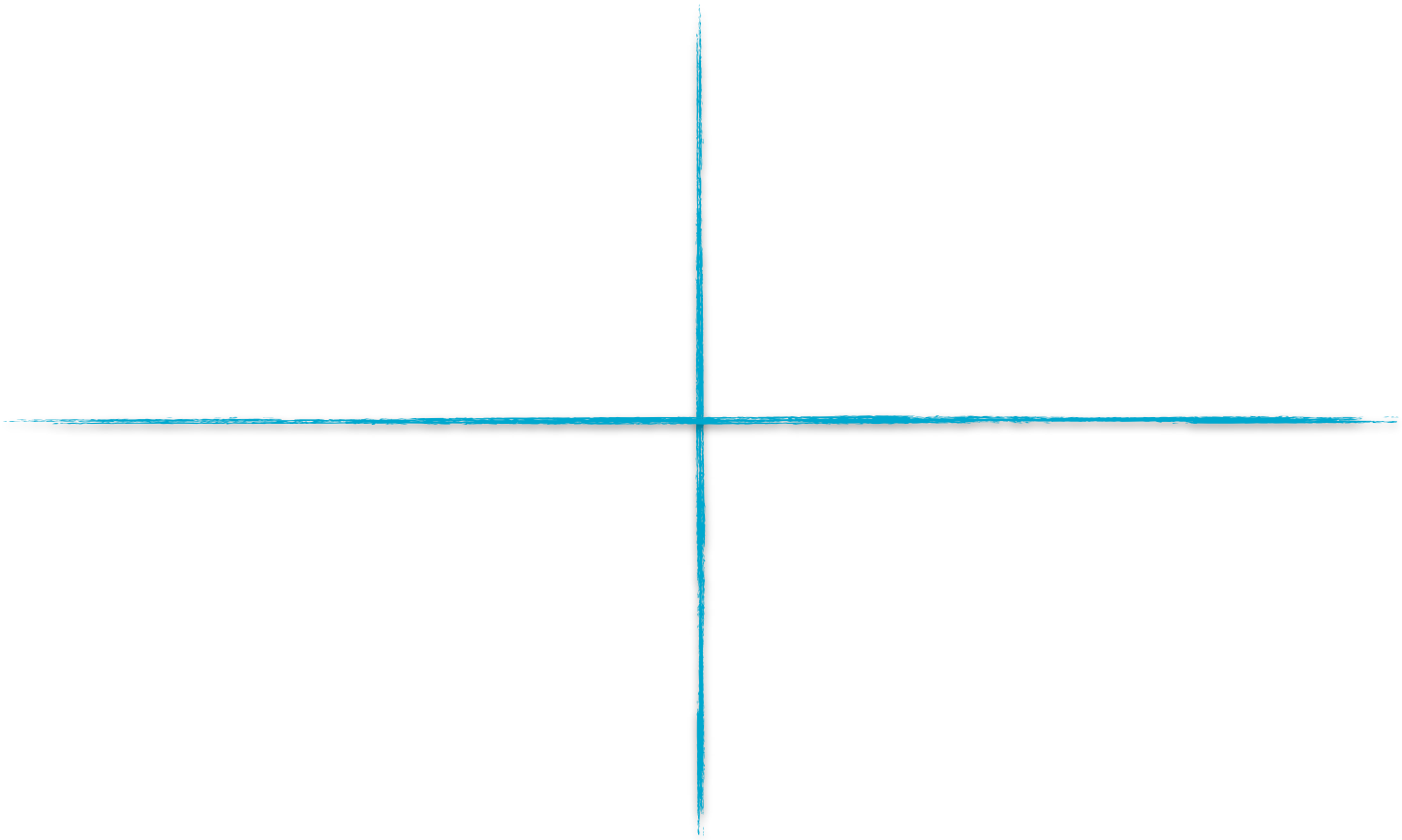
Discharge diagnosis

…

Breakdown of diagnosis reconciliation



4 categories of red ‘flagged cases’



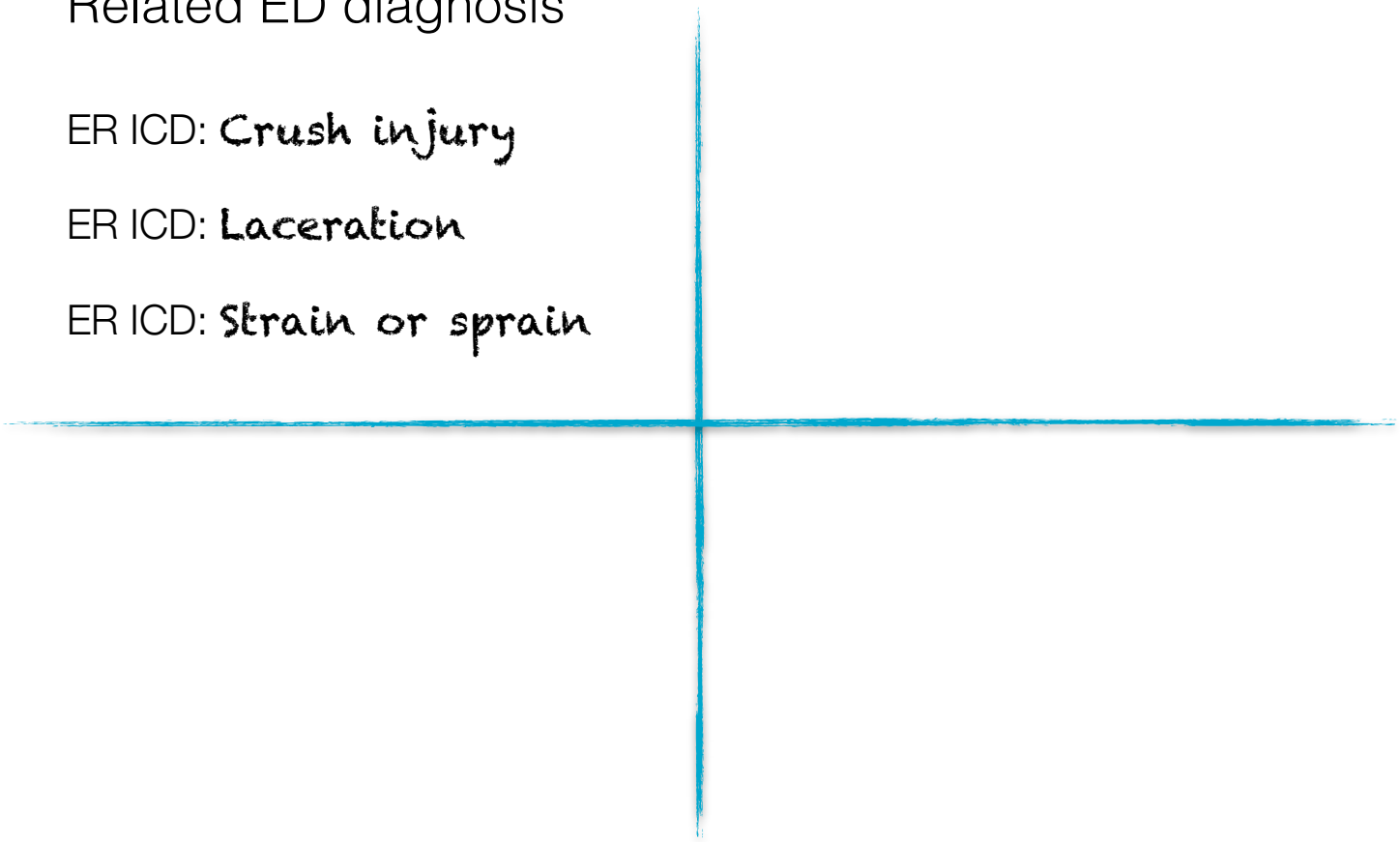
4 categories of red ‘flagged cases’

Related ED diagnosis

ER ICD: Crush injury

ER ICD: Laceration

ER ICD: Strain or sprain



4 categories of red 'flagged cases'

Related ED diagnosis	Unrelated ED diagnosis
ER ICD: Crush injury	ER ICD: Self-harm
ER ICD: Laceration	ER ICD: Congestive heart failure
ER ICD: Strain or sprain	

4 categories of red 'flagged cases'

Related ED diagnosis

ER ICD: Crush injury

ER ICD: Laceration

ER ICD: Strain or sprain

Unrelated ED diagnosis

ER ICD: Self-harm

ER ICD: Congestive heart failure

More general ED diagnosis

ER ICD: Injury of the lower foot.

4 categories of red 'flagged cases'

Related ED diagnosis

ER ICD: Crush injury

ER ICD: Laceration

ER ICD: Strain or sprain

Unrelated ED diagnosis

ER ICD: Self-harm

ER ICD: Congestive heart failure

More general ED diagnosis

ER ICD: Injury of the lower foot.

Missed ED diagnosis

ER ICD: No injury found

ER ICD: Patient did not wait.

Final messages

- Can effectively identify limb abnormalities from radiology reports using NLP+ML
- Robust across different hospital's reports
- Classification results can be reconciled against discharge diagnosis
- Potentially undiagnosed limb fractures can be identified
- Saving doctors time by only reviewing "flagged" cases
- Extend beyond limb abnormalities to cancers and pathology reports

Australian e-Health Research Centre, CSIRO

Dr. Bevan Koopman
Dr. Anthony Nguyen
Mr. John O'Dwyer
Dr. Amol Wagholikar

Royal Brisbane Hospital
Royal Brisbane Children's Hospital
Gold Coat Hospital

Dr. Kevin Chu
Dr. Ibrahim Mahmoud
Dr. Kim Lai
Dr. Gerben Keijzers

Funding by Queensland Emergency Medicine Research Foundation Grant, EMPJ-11-158-Chu- Radiology.
No disclosure.